JC TECH KNOWLEDGE HUB -- TECHNICAL SPECIFICATION SERIES

CAD Standards Library

Standard Operating Procedures for 2D Drafting, 3D Parametric Modeling & GD&T

1. Strategic Importance of Enterprise CAD Standards

Ambiguous technical drawings and unconstrained 3D models cost manufacturing enterprises millions in scrap, tooling modifications, and engineering change delays. The JC Tech CAD Standards Library provides an end-to-end operational protocol that unifies design language across design teams, simulation departments, toolmakers, and external suppliers. Grounded in ASME Y14.5-2018 and ISO 128 specifications, these standards ensure that every dimension, datum reference, and geometric control conveys unambiguous manufacturing intent.

2. Core Pillars of the CAD Standards Framework

- **Standardized Title Blocks & Border Architecture:** Pre-configured sheet formats (A0 through A4) incorporating automated metadata attributes (Part Number, Material, Mass, Finish, Revision, Approval signatures, and Third-Angle projection symbol).
- **Geometric Dimensioning & Tolerancing (GD&T):** Full implementation of Feature Control Frames, material condition modifiers (MMC, LMC, RFS), and Datum Reference Frames to ensure perfect assembly interchangeability without micro-tolerancing.
- **Parametric 3D Feature Tree Hierarchy:** Strict modeling sequence: Base Reference Planes -> Primary Datum Features -> Main Extrusion/Revolve -> Secondary Machining Features (Chamfers/Fillets placed strictly at tree end).
- **Model-Based Definition (MBD / 3D PMI):** Direct annotation of 3D CAD models per ASME Y14.41, eliminating 2D paper drawings in digital enterprise workflows while supporting automated CMM programming directly from STEP AP242 files.

3. Standard Layer & Lineweight Specification (2D Drafting)

Layer Name	Color Code	Lineweight	Linetype	Standard Functional Usage
01_VISIBLE	White / Black	0.35 mm	Continuous	Visible part outlines and primary solid edges
02_HIDDEN	Cyan / Grey 8	0.18 mm	Hidden (DASHED)	Interior cavities, obscured edges, undercuts
03_CENTER	Red (Color 1)	0.18 mm	Center (LONG-SHORT)	Hole centerlines, pitch circles, axes of symmetry
04_DIMENSION	Green (Color 3)	0.25 mm	Continuous	Linear/angular dimensions, leaders, tolerance text
05_HATCH	Yellow (Color 2)	0.18 mm	Continuous	Section cutting plane cross-hatching per material ISO
06_CONSTRUCT	Magenta (Color 6)	0.13 mm	Phantom	Pitch lines, travel boundaries, fixture clearances

4. GD&T Symbol Master Reference & Feature Control Frames

Characteristic	Symbol & Class	Tolerance Zone Description	Primary Industrial Use Case
Position (True Position)	Circle with Cross (Locational)	Cylindrical or planar zone relative to Datums A, B, C	Fastener clearance holes & precision locating dowels
Flatness	Parallelogram (Form)	Zone between two parallel planes (no datum needed)	Gasket sealing surfaces & heat sink mounting pads
Cylindricity	Circle with side tangents (Form)	Zone between two co-axial cylinders	Engine cylinder liners & hydraulic valve spools
Perpendicularity	90 deg Tee (Orientation)	Tolerance zone 90 deg relative to specified datum	Cylinder faces relative to locating baseplate
Profile of a Surface	Semi-circle (Profile)	3D envelope enclosing the true CAD nominal surface	Castings, automotive exterior panels, turbine blades
Total Runout	Double arrow connected (Runout)	Full 3D indicator composite runout when rotated 360 deg	Transmission drive shafts & high-speed rotor journals

5. Parametric 3D Modeling Best Practice Rules

- **Rule 1: Fully Constrain Every Sketch:** Under-constrained sketches (blue lines) are strictly prohibited. Every profile must be dimensionally or relationally constrained to origin planes.
- **Rule 2: Never Create Fillets Inside Primary Sketches:** Keep base sketches strictly orthogonal. Apply blend radii as 3D solid edge features to prevent downstream sketch regeneration corruption.
- **Rule 3: Establish Skeleton Master Models for Assemblies:** Use top-down skeletal geometry to control mating clearances across multi-component sub-assemblies.

6. Enterprise Engineering Change Protocol (ECR / ECO)

ECR Review
Formal Change Request Gate

CAD Update
Parametric Revision & Check

Simulation Audit
FEA/CFD Integrity Re-check

PLM Release
Authorized ERP Drawing Dist.

Download Ready-to-Use CAD Template Packs

JC Tech provides verified drawing and model templates for SolidWorks, CATIA V5, Creo Parametric, Siemens NX, and AutoCAD with embedded standard layers, title blocks, and tolerance tables.